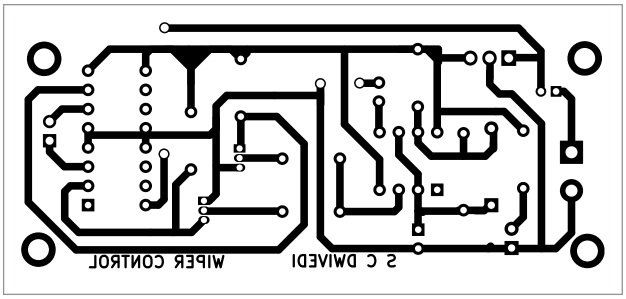

Fig. 3: Actual-size PCB of the circuit

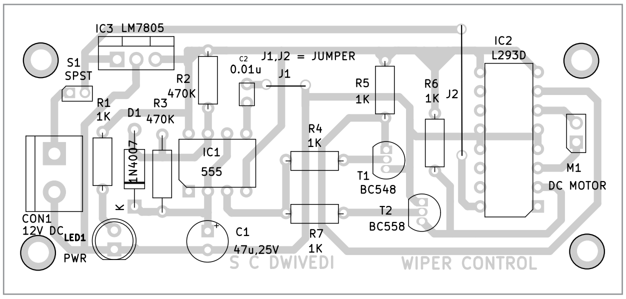

Fig. 4: Component side of PCB

the circuit, keeping the wiper motor inactive. When rain begins, switch S1 on to activate the circuit and initiate the wiper motor's rotation to clean droplets from the vehicle's windscreen.

The wiper rotates clockwise and anticlockwise based on the signal received at pins 2 and 7 of the L293D. These signals are provided by the output of timer 555 (IC1) through transistors T1 and T2. The L293D output is connected to a geared motor to which the wiper is fitted. Enable pin 1 is connected to a high voltage of 5V (logic 1) to enable the driver. The wiper motor receives power and control signals to operate the wipers.

The L293D has 16 pins in total, as follows:

Pin 1 is Enable 1, which turns on/off the first H-bridge. Pin 2 is input 1 (1A), which is the first input for motor 1. It is connected to the emitter of the PNP transistor T2. Pin 3 is output 1 (1Y) for motor 1. Pins 4 and 5 are grounded. Pin 6 is output 2 (2Y) for motor 1. Pin 7 is input 2 (2A), which is the second input for motor 1. Pin 8 is connected to 12V to power the wiper motor.

These first eight pins of the L293D are used to control the first DC geared motor to rotate the wiper.

Similarly, there are two inputs and two outputs to control the second DC motor, though it is not used here. For more details, refer to the datasheet. See the table on previous page for the direction of motor rotation.

Construction and testing

An actual-size, single-sided PCB layout for the automatic wiper control for three/four-wheelers is shown in Fig. 3, and its component layout in Fig. 4. After assembling the circuit on the PCB, enclose it in a suitable box. Fix LED1 and CON1 on the front side of the box. Switch S1 should also be fixed on the front side of the box for convenient on/off control of the circuit. LED1 should glow when S1 is turned on after connecting a 12V battery or adaptor across CON1 to enable the circuit.

After assembling the circuit, enclose it in a suitable box and install the wiper on the motor shaft. Position the wiper to cover the maximum glass area when rotating. The automatic wiper control for three/four-wheelers is now ready for use. EFY

Bonus. You can watch the video of the tutorial of this DIY project at *https://www.electronicsforu.com/videos-slideshows/live-diy-automatic-wiper-control-for-three-four-wheeler*

PARTS LIST

Semiconductors:	
IC1	- 555 timer
IC2	- L293D motor driver
IC3	- LM7805, 5V voltage regulator
T1	- BC548 NPN transistor
T2	- BC558 PNP transistor
D1	- 1N4007 rectifier diode
LED1	- 5mm LED (green/red)
Resistors (all 1/4-watt, ±5% carbon):	
R1, R4-R7	- 1-kilo-ohm
R2, R3	- 470-kilo-ohm
Capacitors:	
C1	- 47µF, 25V electrolytic
C2	- 0.01µF ceramic disk
Miscellaneous:	
CON1	- 2-pin connector
M1	- 12V geared DC motor
	- 12V battery

S.C. Dwivedi is an electronics enthusiast and circuit designer at EFY